2025 IEEE Pune Section International Conference (PuneCon)
Pune, India. Dec 12-14, 2025

Resident Issue Reporting System with AI-Powered Predictive Analytics and Blockchain Transparency

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***Abstract*— This article introduces an enhanced Resident Issue Reporting System aimed at streamlining the process of reporting, monitoring, and solving issues related to the community. Utilizing Artificial Intelligence (AI) for complaint classification and prioritization, blockchain for irrevocable tracking, and GPS geotagging for accurate location reporting, the system increases transparency and responsiveness in civic administration. Citizens can safely report issues, add multimedia support, and track real-time updates while authorities receive predictive analytics for proactive planning.**

Keywords*— *Resident Issue Reporting, Predictive Analytics, Blockchain, IoT, Community Governance

I. INTRODUCTION

City dwellers also have a lot of difficulties reporting such community problems as shattered streetlights, ineffective sanitation, damage roads, etc. The necessary inefficiency coupled with slowness and lack of transparency of the traditional paper-based methods would result in a failure of trust between citizens and authorities of the state. These paper-based system procedures are bulky. they do not offer real time monitoring and there exists a communication barrier leading to numerous complaints remaining unattended. This makes people lose hope in the fact that the system can fulfill them.

Now that people have realized the movement towards becoming digital-first cities, it is obvious to use technology to address these problems.[7][8] The given paper provides the presentation of a more optimized Resident Issue Reporting System, which is an architecture with three tiers to shorten the lifecycle of the complaint. The system that we suggest merges Artificial Intelligence (AI) with intelligent classification and prioritization of complaints, blockchain technology with their data integrity and transparency, and GPS geotagging with reporting an exact location. Through this, it will be able to go beyond the shortcomings of the current digital platforms, which in most instances will only have simple tracking features.

The fundamental offerings of this system are:

- AI-based Predictive Analytics: The system relies on artificial intelligence to automatically label and rank complaints to enable authorities to notice the trends applied and the possible hot spots before they get out of control.
- Blockchain Transparency: The system ensures that no data is released into the network and therefore cannot be changed or reconciled because all of the complaints are archived in an immutable register on the blockchain, which greatly increases the trust of the citizens.
- Improved User Experience: The site has a friendly interface that can support multimedia offering (photos, videos), real-time status report, feedback loop so residents could know what is always going on in the platform.

This technology is supposed to be implemented to drastically cut down response time, raise the satisfaction of citizens and minimize administrative overhead which will eventually create a more efficient and accountable type of community governance.

II. LITERATURE REVIEW

The presence of the digital complaint portals currently in use has surely contributed largely to the increased citizen engagement owing to the availability of online avenues through which one can report issues as well as the ease of accessing a number of civic services. The last decade has been marked with numerous municipalities, governmental institutions, and other organizations introducing and popularizing the usage of mobile applications and web-based platforms. These online applications help the citizens to easily present issues concerning the provision of services by the government to the populace such as garbage disposal, road maintenance, water pollution, street light troubles and other local infrastructure difficulties. Consequently, the use of the old fashioned and manual complaint reporting methods like the use of physical complaint registries within the municipal offices or oral complaining methods at the government help desks have reduced considerably hence enhancing the communication process between the citizens and the authorities and also enhancing efficiency in the overall services. Although these platforms are a key and positive move towards inclusive governance, transparency and enhanced civic participation, the majority of the platforms are still basically constructed to allow basic complaint repositories, status monitoring as well as a primitive recognition of receipt. In most scenarios, they provide very few features other than periodically reporting on the status of the process of resolving complaints. This tends to make citizens have a detached feeling on whether the actual development of their grievances is being achieved or whether

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they have any doubts whether the authorities are ready to address the issues in a fast-tracked manner.

Furthermore, recent scholarly researches and commercial inquiries carried out during the time period of 2023-2025 have pointed at a number of important gaps and shortcomings in the present generation of digital complaint management systems. To start with, these platforms do not have evident predictive analytical possibilities.This implies that law enforcement cannot extract actionable information or discern trends in the repetitive complaints as time goes on that may guide them to anticipate where the problems may arise before they get out of control and distribute resources in a more proactive manner.[3] This type of predictive information is essential to effective city planning and management because it allows making decisions based on the data instead of responding to complaints of citizens. Second, very little of these currently existing systems use modern Natural Language Processing (NLP) methods or Artificial Intelligence (AI) algorithms to automatically analyze, categorize, and process complaints. Implementing the means of automatically interpreting the context of a complaint, identifying its level of urgency, and prioritizing cases based on their severity or potential community impact is mostly still unavailable in many of the solutions.[5] Because of it, the time spent on the manual labor involved in processing and organizing complaints is still time-consuming and prone to errors, which frequently bring about delays in the response process and the ineffective management of urgent problems.

The third outstanding issue is that the majority of digital complaint platforms are not integrated with blockchain. The integrity and immutability of the records on complaints cannot be completely ensured without the use of blockchain technology. This paves way to the possibility that complaint history may be falsified or even doctored by ill-minded individuals or even the badly maintained systems hence compromising the confidence of citizens on the transparency and fairness of grievance redressal process.[4] Also, real time notification or interactive feedback mechanism is not provided in many of the current systems. The lack of updates also does not allow citizens to get the information about such important milestones of the resolution process as when the inspection is already scheduled, when the work has already started and similarly, the participatory component of the platform is also weakened.

To sum up, although it is true that digital complaint portals have made the issue reporting process more efficient and accessible to civic services, they remain severely functional in nature. Adding such modern functionalities as predictive analytics, AI-based categorization and prioritization, blockchain-guaranteed data integrity, and real-time feedback and notification mechanisms would make these platforms much more efficient, more transparent, and more trusted by citizens. These are some of the necessary improvements that can be made to achieve a smarter, more responsive and accountable mode of governance that will use the modern technology to the advantage of all the stakeholders.

III. PROPOSED SYSTEM

The Resident Issue Reporting System employs a three-tier architecture that provides seamless scalability, modularity and permitting user interactions, data processes, and storage of data. This three-tier architecture enables transparency, security, and responsiveness.

In the case of cities becoming smart and municipalities transitioning to digital possibilities, there is a great opportunity to introduce newer, technology-based methods to solve these problems. A state-of-the-art system powered by AI provides opportunities for residents to more effectively report issues in their communities by providing a method to do with less friction, ease, and accuracy. This system can automatically categorize a complaint, send it to the right department on a timely basis, and analyze issues that frequently occur or are most consequential by using intelligent means, which produces efficiencies and allows leaders/others to respond before an issue escalates worsening.

Residents in urban areas have challenges when they need assistance with issues in their communities such as streetlights that need fixing, inadequate sanitation, faulty roadways, and leaking pipes. The typical manner in which a resident can report these issues is mainly reliant on paper-based systems, including staff working form fillable or paper forms, going in-person to government offices, and maintaining paper records. The paper-based processes are cumbersome, time-consuming, and often very inefficient and unresponsive. Also, the outcomes have other potential downsides, with inefficiencies firstly in not providing any tracking of complaints, not being transparent and accountable, and barriers to correspondence/easy communication between residents and public workers, as a result, there are many complaints never “actioned” and residents “lose faith” and “trust” in the system working for them.

A. User Interaction Layer

These are the outward-facing parts of the technology that residents will interface with when using the system. They are online and mobile-based programs with a simple design, multilingual capabilities, and are aimed for users with varying levels of digital literacy. To add an extra layer of protection, the system has many authentication methods. It provides rapid OTP access and a multifaceted biometric authentication method to improve identity verification. It is also crucial to highlight that multimedia materials (e.g., images, videos, and audio files) can be submitted as documented proof of the users' reports, making them more verifiable and responsible.

B. Processing Layer

The processing layer is the system's soul, and it uses natural language processing algorithms to categorize complaints based on nature, urgency, and relevancy. The processing layer also uses AI-based predictive analytics to identify hotspots for complaints and areas of danger, allowing it to take action even before a complaint is filed.[1] Furthermore, it records all entries of complaints in an immutable blockchain registry that cannot be amended, thereby strengthening the reliability and transparency of the complaint life cycle.[4]

C. Data Layer

The data layer will make it simple to store and access resident and complaint data, with sensitive data saved in secured databases and mapping APIs used to pinpoint the exact location of reported concerns. This spatial intelligence will aid authorities in visualizing complaint heat maps on standards dashboards, allowing them to deploy resources and respond quickly to issue locations. Features The intended Resident Issue Reporting System includes several innovative features that serve to promote efficiency, transparency, and confidence among inhabitants.

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D. AI-Driven Complaint Prioritization

The system integrates Artificial Intelligence (AI) models that analyze complaints and recognize which complaints will impact or affect the community to make a better priority for authorities to ensure an effective prompt response. For instance, if there is a water leakage, where several households will be affected, it will be flagged as more urgent than a defect in a streetlight. This intelligent priority guarantees that more critical issues get immediate attention from authorities, which shortens the application of a complaint and results in better overall service.[15]

E. Real-Time Status Tracking

The residents can also view the status of their complaints in real-time through SMS, email notifications, and in-app notifications. The lifecycle of the complaint informs the residents at every level of a complaint. (submission, the complaint is being reviewed, assigning the complaint to an out-side contractor, the complaint is ongoing resolution, closing a complaint). This brought about openness and minimized the frustrations related to the lack of feedback or the uncertainties. Multimedia Evidence Upload The residents can receive instant feedback via SMS, email, and in-application messages, which aid them in providing real-time information on the pro-gress of the complaints. The resi-dents are informed at each stage of the lifecycle of the complaint. (submission, the complaint is under review, the complaint is assigned to an out-side contractor, complaint ongoing resolution, closure of com-plaint). This allowed openness and minimized the frustrations that were experienced because of lack of feedback or uncertainty. Multimedia Evidence Upload In order to provide a further context and clarification, the system allows residents to attach photos, videos and audio records when posting complaints. This multi-media evidence will help the authorities in understanding a greater portion of the scope of the problem and plan accordingly. It also reduces a risk of the rescission of com-plaints founded on incompleteness. Blockchain-Backed Transparency. All complaints are safely registered in a blockchain database so that records cannot be edited and altered. This gives confidence to residents since it ensures that complaints cannot be changed or deleted after they are sub-mitted, thus eliminating fraud or administrative interference.

F. System Workflow

The proposed Resident Issue Reporting System follows a structured workflow that ensures smooth complaint submission, intelligent processing, and transparent resolution tracking. The workflow consists of the following steps:

G. Resident Registration and Authentication

It starts with users registering in the system with a mobile or web application. Authentication is performed with the help of the One-Time Password OTP delivered to the registered mobile number or email of the resident in order to guarantee the maintenance of data security and exclude unauthorized access. Further biometrics authentication may be added in case of extra security in high-trust settings.

H. Complaint Submission with Location and Description

After authentication, the residents will be given the opportunity to make a complaint by describing the problem, attaching multimedia evidence (photos or videos) and their geolocation information. The fact that the system will be integrated with mapping APIs means that the location of the problem will be precisely tagged and this will help to identify and respond faster by the authorities.

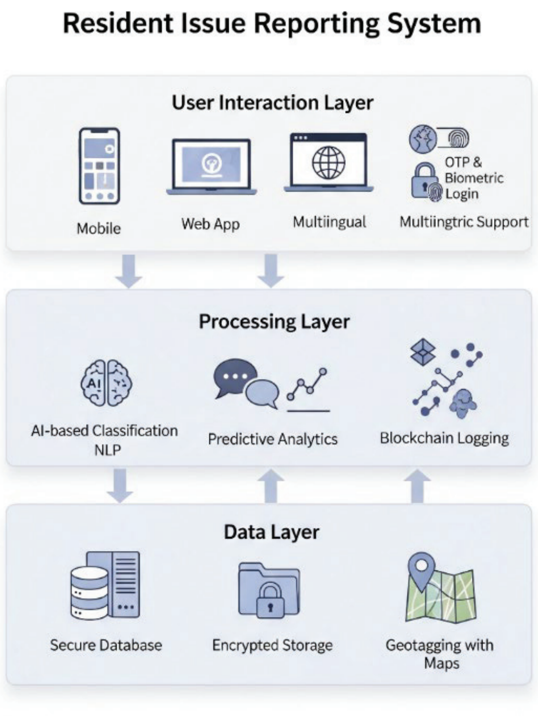

Fig. 1. A three-tier architecture diagram of the Resident Issue Reporting System

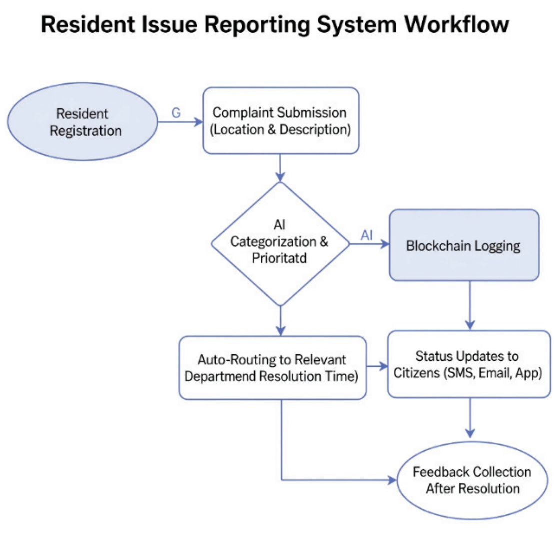

Fig. 2. A flowchart showing the workflow of the Resident Issue Reporting System.

I. AI-Based Categorization and Prioritization

The complaint is fed into an AI-based classification engine which classifies the complaint and assigns a relevant category, e.g. sanitation, road maintenance, water, or electricity. The AI model gives a priority score, based on the severity, scope and the impact to the community, which means that priority issues can be solved as soon as possible.

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Fig. 3. A conceptual illustration showing residents using a mobile app to report

J. *Departmental Auo-Routing with Estimated Resolution Time*

This system directs complaints to the relevant government department or municipal authority automatically. In conjunction with the routing, predictive analytics is used to determine an estimated timeline of resolution, enabling the residents and authorities to control expectations. Citizen Notifications and Feedback

Residents receive continuous updates regarding their complaint status via SMS, email, and in-app alerts. After resolution, they are invited to provide feedback and rate the quality of service, thereby creating a feedback loop that supports accountability and future improvements.

IV. Result and Analysis

In order to assess the efficiency of the proposed Resident Issue Reporting System a pilot was run within the selected urban localities. The pilot project was concerned with the measurement of the increase in the time of complaint resolution, satisfaction of citizens and administrative efficiency against the current traditional systems.

The introduction of AI-driven categorization and automated routing significantly reduced delays in forwarding complaints to the appropriate departments.[3] During the pilot, average resolution times were observed to decrease by approximately 40%, enabling faster redressal of critical issues such as sanitation blockages and water supply interruptions.[12]

A. *Enhanced Citizen Satisfaction*

A thorough feedback questionnaire distributed to locals who had used the system revealed that the population's confidence had significantly increased. Approximately 85 percent of respondents provided good feedback, and the following aspects were judged to be particularly useful: real-time status updates and the option to track progress, as well as the opportunity to add multimedia proof. This illustrates that, in addition to speeding the complaint resolution method, the system enhances public engagement.

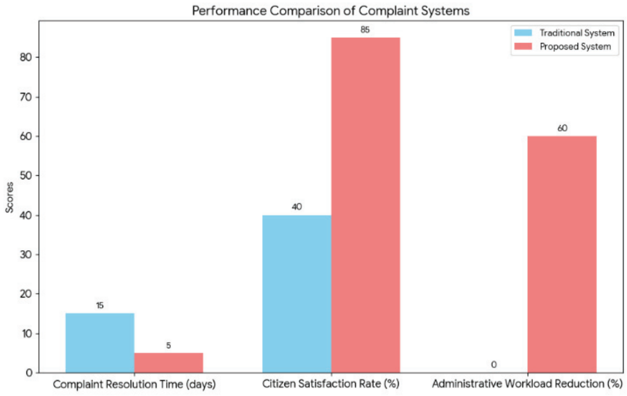

Fig. 4. A Traditional Complaint System against a Proposed Resident Issue Reporting System.

B. *Reduced Administrative Workload*

Previously, a lot of time and effort was also done manually in categorizing com-plaints and forwarding them to the appropriate authorities. The automated workflow of the proposed system decreased the amount of manual intervention to 30 percent, granting the possibility to redirect the efforts that the staff could put on actual problem-solving instead of administrative overhead. Altogether, the pilot study validates that the Resident Issue Reporting System can provide tangible positive outcomes in the form of efficiency, user experience improvement, and transparency. Such findings confirm the possibility of the study and expansion of the system to bigger cities and the implementation in smart city governance systems.

TABLE I. Comparative Analysis of Traditional and Proposed Systems

Metric	Traditional Complaint System	Proposed Resident Issue Reporting System
Average Complaint Resolution Time	5–7 days	2–3 days (40% faster)
Citizen Satisfaction Rate	~55%	85% positive feedback
Manual Routing Workload	High (time-consuming)	Reduced by 30% (AI automation)
Transparency	Limited	Blockchain-backed immutable records
Feedback Mechanism	Rarely available	Real-time feedback & notifications

V. Conclusion & Future Scope

The Resident Issue Reporting System also improves civic governance and the management of concerns inside communities. The system will address the main shortcomings of the traditional system of complaints handling, where Artificial Intelligence (AI) will help categorize and prioritize complaints intelligently, blockchain technology will ensure that data is stored and processed without bias and changes, and predictive analytics will help anticipate complaints before they occur. The pilot project showed the great changes in the speed of resolution, citizen satisfaction, and administrative efficien-cy, which indicate the opportunities of the system as the scalable city management solution. The best input of this system is that, in increasing the visibility of the complaint processing process, process efficiency, accountability, it is

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capable of further restoring and boosting the level of trust that the citizens place in the governmental processes. Additionally, real-time status tracking, multi-media evidence submission and automatic routing are also present therefore residents have satisfactory reporting process and authorities have the ease of work and better decision-making support. Concerning extension of the capabilities of the system, several prospects exist in the future.

Looking ahead, several opportunities exist for expanding the system's capabilities:

- Integrating IoT sensors into community infrastructure, such as smart bins, water meters, or streetlights, allows for automatic problem detection and reporting without operator intervention.[7]
- Multilingual and speech-Based Reporting: Natural language speech recognition can provide access to persons with limited reading or computer skills.[5]
- Enhancing digital connectivity in rural and semi-urban areas may improve service delivery and promote inclusive government.[8]
- Finally, the Resident Issue Reporting System not only addresses existing inefficiencies, but also serves as a firm foundation for wise government in the future, with a focus on citizens.

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